MOBILE PROGRAMMING MIDTERM · TEAM BLUEBERRY

NewUU Campus Updates

One feed for room changes, cancellations and deadlines, plus a finder for free study rooms.

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THE PROBLEM

Campus updates are scattered across three places, and none is quick to check

Telegram

One channel per course. A room change is buried under chat and easy to mute.

Hero Study

Slow and confusing. It cannot mark a room change or a cancellation.

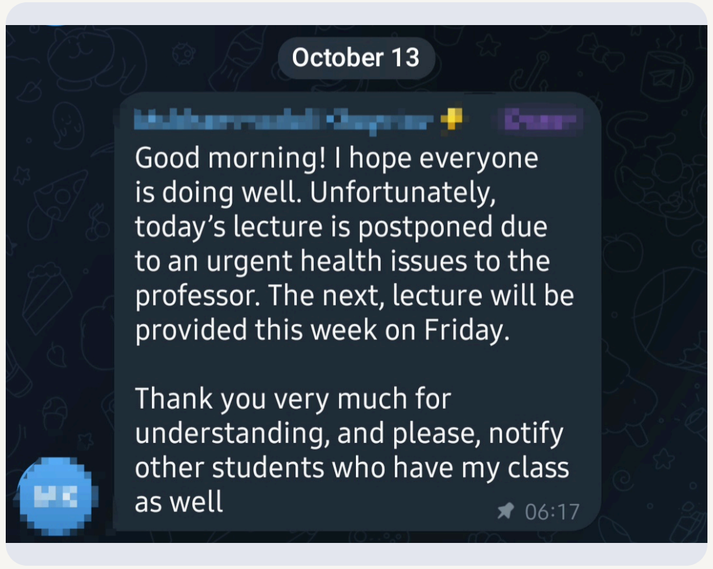
EduPage

Holds every room’s timetable, but finding a free room means reading it room by room.

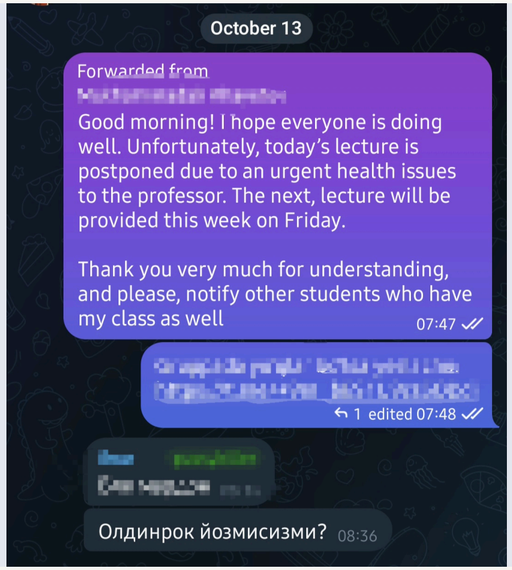
Result: students go to the wrong room, show up to cancelled classes, miss deadlines, and can’t find a quiet place to study.

EVIDENCE

A postponement posted at 06:17 still hadn’t reached a student by 08:36



Oct 13, 06:17: the professor asks students to “notify other students who have my class as well.”



Forwarded at 07:47. At 08:36 a student asks: “Couldn’t you have written earlier?”



The library during the day: nearly every seat is taken.

TARGET USERS AND SCOPE

Two roles: students read, professors post

Students (primary)

NewUU undergraduates with 4 to 6 courses. Need one place for today’s changes and deadlines, and a free room to study.

Professors (secondary)

Teaching staff with 1 to 4 courses. Need a fast way to post to one course and know students saw it.

Scope: NewUU students miss class updates and can’t find free study rooms because campus information is scattered across three different places.

APP CONCEPT AND MAIN FLOW

Open the app and see, on one screen, what changed today

Professors post short typed updates (room change, cancellation, deadline, note). Students see them in one feed, filtered to their courses. A free-room finder uses the EduPage timetable.

1
Sign in as a student

2
See today’s classes and updates

3
Tap a card to read the new room or time

4
Open the course for its history

Professor flow: sign in, tap +, pick course and type, fill the fields, post. Under 30 seconds.

SCREENS

Seven screens, one job each

1. **Login:** sign in, pick the role
2. **Today feed:** today’s classes and updates
3. **Course detail:** one course’s history
4. **Deadlines:** all deadlines, soonest first
5. **Post update:** professor posts in 30 s
6. **Free rooms:** rooms free right now
7. **Room schedule:** how long a room stays free

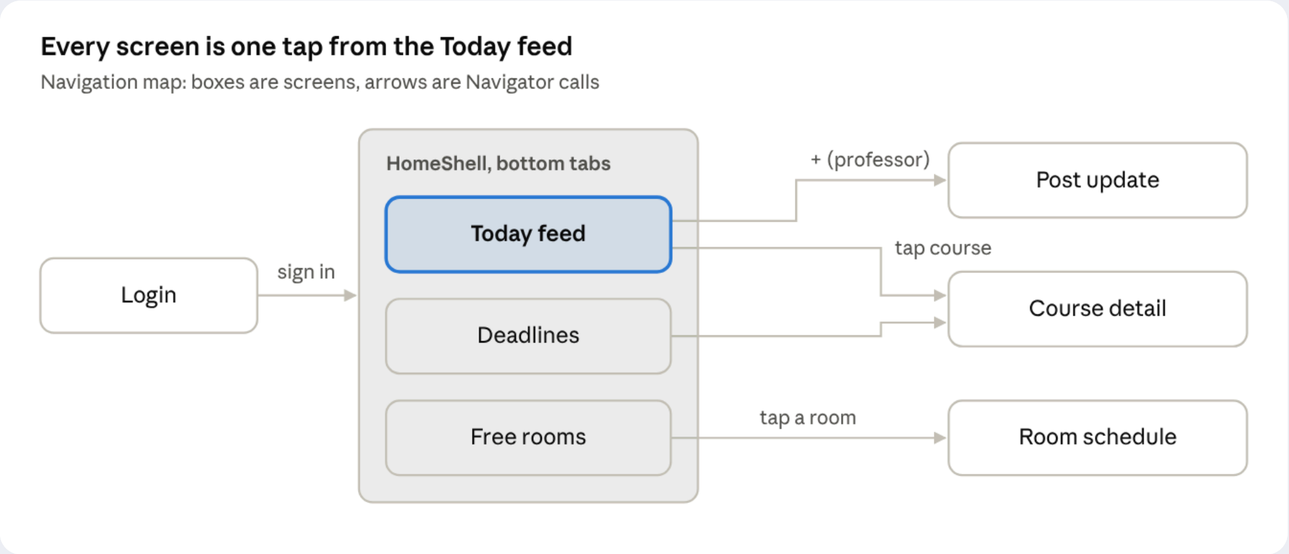
OUR DESIGN

The full mobile UI is in Figma

Open the design in Figma

FLUTTER PLAN: NAVIGATION AND STATE

Named routes move between screens; Riverpod holds shared data



- **Login** uses pushReplacementNamed, so Back skips it
- **HomeShell**: BottomNavigationBar + IndexedStack; tab index in setState
- **Post update** returns the new update with Navigator.pop

Riverpod: ProviderScope at the root. updatesProvider and authProvider are NotifierProviders; three screens ref.watch the same updates, so the list is lifted out of any one screen.

FLUTTER PLAN: WIDGETS AND FORMS

Lists are builders; two screens validate input

Screen	Main widgets	Form and validation
Login	Column, SegmentedButton, TextFormField ×2	Form + GlobalKey; email ends in @newuu.uz; password ≥ 6
Today feed	ListView.builder, ExpansionTile, FAB (professor)	None
Course detail	TabBar, TabBarView, ListView.builder	None
Deadlines	FilterChip row, ListView.builder	None
Post update	DropdownButtonFormField, ChoiceChip, date picker	Course required; room like B-204; date not past; message ≤ 200
Free rooms	DropdownButton, ChoiceChip, GridView.builder	None
Room schedule	ListView.builder of Rows, Stack for the now line	None

DATA MODEL AND OUT OF SCOPE

Four Dart classes, sample data for now

AppUser

id, name, email, role (student or professor), courseIds

Course

id, code, title, professorName

CourseUpdate

id, courseId, type, message, postedAt, and nullable newRoom, affectedDate, deadlineTitle, dueAt; seenCount

TimetableSlot

courseId, room, building, weekday, startMinute, endMinute; isBusyAt(time)

Out of scope (after Week 7): backend and database, Hero Study and EduPage integration, push notifications, a real “seen by” count.

WHO DID WHAT

Five members, one part each

Member	Part
Hazratov Mardon (team lead)	Problem research and evidence
Umarov Istamjon	User flows and wireframes
Usmonov Muhammad Asadxon	Data model
Xushvaqtov Elshod	Presentation
Erkinov Sardor	Flutter plan and Room schedule screen

Thank you. Questions?